

Resource use and climate evidence Teacher notes

LAUNCH Humanities | Sum1 W1 | SoW slot 27 | 40 minutes in one session

Teach the exact weekly objective below using the nine-stage lesson. Select one route for each pupil, receive their response in an accessible form and agree a next step. These notes include the worked explanations, answers and evidence expectations.

Objective and evidence

Explain resource use and climate change using online data.

F: source annotation, evidence selection and questioning.

- Identify the source, date, unit and geographical scope of a dataset.
- Use two figures to describe an emissions pattern.
- Link resource use to climate change and state one limitation.

Preparation and access

Use the embedded historical snapshot; pupils do not need live web access. Pre-teach MtCO₂e and territorial. Explain that the source uses rounded headline totals. Assess the pupil's evidence and reasoning, never demand a personal political belief.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

A fall in emissions is not a fall of the same number of degrees. These are net annual UK territorial emissions, not atmospheric concentrations or the whole UK consumption footprint.

Vocabulary to teach first

resource (Tier 2): something people use to meet a need.

evidence (Tier 2): information used to support a claim.

territorial (Tier 3): occurring within a defined territory; here, the UK.

emissions (Tier 3): gases released into the atmosphere.

MtCO₂e (Tier 3): million tonnes of carbon dioxide equivalent; combines greenhouse gases using their warming effect.

limitation (Tier 2): a reason evidence cannot answer every question.

A sustainability analyst checks data provenance before advising a council. Use a figure, a geographical boundary and a limitation to make your advice trustworthy.

Teaching sequence

Stage	Minutes	Teacher action and pupil task
Title	0	Name one resource used to provide energy.
Arrival	4	Choose one route and answer four retrieval questions.
Starter	4	Identify what you would check before trusting an online chart.
I Do	5	Watch how I audit the source and describe the figures.
We Do	5	Choose the statement supported by the emissions dataset.
I Do 2	4	Watch how I link resource use, emissions and warming without overclaiming.
We Do 2	4	Repair an overclaim by adding a geographical limitation.
Independent	10	Explain resource use and climate change using two figures and one limitation.
Exit	4	Give one supported claim and one thing the dataset cannot tell us.

I Do worked explanation 1

Who: the UK Department for Energy Security and Net Zero (DESNZ). Published: 5 February 2026.
 Measure: net UK territorial greenhouse gas emissions, in MtCO₂e. The rounded totals fall from 384 in 2023 to 373 in 2024: a fall of 11 MtCO₂e, about 3%. This tells us about emissions inside the UK, not the world total.

I Do worked explanation 2

People use fossil fuels for electricity, heat and transport. Burning them releases carbon dioxide, a greenhouse gas that contributes to warming. DESNZ attributes much of the UK fall in 2024 to lower gas and coal use in electricity supply and industry. But this UK dataset alone cannot establish global temperature change or count all emissions linked to imported goods.

We Do checks

Which statement is supported by the rounded figures?

384 – 373 = 11 MtCO₂e. This is a UK emissions measure, not world emissions or temperature.

Choose the sound limitation.

Territorial means inside the UK. Overseas supply-chain emissions associated with imported goods are outside this measure.

Arrival questions and answers

Use one route only. Hints sit beside questions in the lesson and pupil resources. Accept equivalent meaning in the pupil's response mode.

Supported

1. Which is a fuel resource: coal or a chart title?

Answer: Coal.

2. Does emissions mean gases released or distances travelled?

Answer: Gases released.

3. Which is larger: 384 or 373?

Answer: 384.

4. Which helps check online evidence: its author and date or only its colour?

Answer: Its author and date.

Standard

1. Name one fossil fuel used for energy.

Answer: Coal, oil or natural gas.

2. Why record the date of an online dataset?

Answer: To know the time period and which release is being used.

3. Find the difference between 384 and 373.

Answer: 11.

4. Does a UK figure automatically describe the whole world?

Answer: No; the geographical coverage differs.

Stretch

1. How can burning a fossil fuel affect climate?

Answer: It releases greenhouse gases, including carbon dioxide, which contribute to warming.

2. Name two checks before using a chart as evidence.

Answer: Any two of author, date, unit, geographical scope or method.

3. A value falls from 384 to 373. Estimate the percentage fall.

Answer: About 2.9%, or 3% rounded.

4. Why does a fall in annual emissions not prove climate change has stopped?

Answer: Annual emissions remain positive; greenhouse gases accumulate, and a UK series is not the global climate record.

Independent task and exit

Supported

Use the frame: “DESNZ reports UK territorial emissions of ____ in 2023 and ____ in 2024, measured in _____. Using less _____ can reduce emissions. Greenhouse gases contribute to _____. This dataset does not show _____.” Say it, write it or direct an exact scribe.

Exit: Give the 2024 UK total and one missing answer.

Expected evidence: 373 MtCO₂e, rounded; the table cannot give world emissions, local air quality or a temperature change.

Standard

Write or present a short geographical explanation: identify the source and date; compare two figures; explain a resource-use link to emissions and climate change; end with one limitation of the dataset.

Exit: Give one quantified pattern and one geographical limitation.

Expected evidence: 384 to 373 MtCO₂e from 2023 to 2024; territorial UK data omit overseas emissions from imported goods.

Stretch

Evaluate the claim: “Falling UK emissions show resource choices can matter, but do not prove the climate problem is solved.” Use two figures, a causal explanation and a geographical limitation. Distinguish emissions from temperature.

Exit: Explain why falling UK emissions and continuing global warming can both occur.

Expected evidence: UK annual emissions still occur, gases accumulate, and the global climate is affected by emissions beyond the UK.

Use the feedback response loop

Provide Space, invite Voice, genuinely receive the response as Audience, then agree Influence. A button records the agreed step; it does not itself prove learning. Keep each stage's response separate. In I Do stages the teacher models without a pupil-response panel. Use the next stage to check understanding.

Record the support used. If a pupil declines, consider regulation, communication, demand, relationship and timing; offer another route or revisit. Do not make completion of a digital panel a condition for access to the next task.

Accreditation caveat from the SoW

UAS means unit Award Scheme: match evidence to the exact selected unit and its requirements before submission. The plans do not identify approved unit codes.

Policy wording and source checks

The quotations below use the supplied Feedback and Marking Policy 2025/2026 Pilot, Issue 1. Body section numbers are used because the contents page differs.

Space

“Regulation first. A dysregulated student cannot respond to feedback. Co-regulation and time take priority over the technical content of the feedback.”

“A trusted key adult or familiar teacher.”

“Time to think. Student is given enough time to form a response”

“A clear, low-cognitive-load environment. Predictable lesson structure. Communication-friendly displays.”

“Permission to refuse; refusal is treated as information. Adults consider regulation, communication, task demand, relationship and timing, then re-offer feedback through an alternative modality where appropriate. Patterns of refusal inform APDR, regulation planning and curriculum access.”

“Establish Space at the start of the lesson - regulation, key adult, success criteria visible, no public comparison.”

First five quotations: body §5, PDF page 8. Final quotation: body §14, PDF page 14.

Staff codes

VF = Verbal feedback given here · WS = Worked with support · I = Independent · NS + ... = Next step one specific thing · E = Evidence on EFL · R = Responded loop closed (Audience has happened) · // = Self-edit point · ? = Read this back to me

EFL capture at Exit

“30-second clip, photo, or one-line context note. Central tags only. Listen / look before tagging (Audience).”

Feedback Policy body §14, PDF page 14. Use an individual response, not an undifferentiated group outcome.

Content sources

DESNZ, Final UK greenhouse gas emissions statistics: 1990 to 2024. Statistical release, 5 February 2026, pages 1, 3, 6 and 13. Rounded headline net territorial totals: 1990 790, 2023 384, 2024 373 MtCO₂e. 1990 = 373 + stated fall of 417. Crown copyright, Open Government Licence v3.0.

<https://www.gov.uk/government/statistics/final-uk-greenhouse-gas-emissions-statistics-1990-to-2024>

Met Office, Causes of climate change. Primary meteorological explanation of fossil-fuel use, greenhouse gases and warming.

<https://www.metoffice.gov.uk/weather/climate-change/causes-of-climate-change>